

ho tutto cose anche su vettori

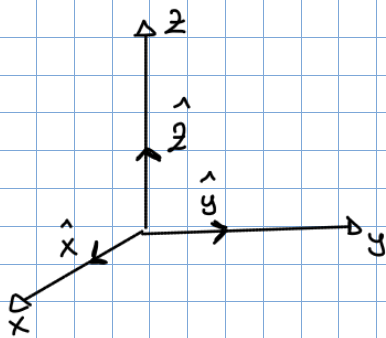
$$\vec{v} = (v_x, v_y, v_z) \quad \text{vettore}$$

$$K \vec{v} = (K v_x, K v_y, K v_z) \quad \text{moltiplicazione scalare} \quad K = \frac{1}{|\vec{v}|}$$

$$|\vec{v}| = \sqrt{v_x^2 + v_y^2 + v_z^2} \quad \text{modulo}$$

$$\hat{v} = \frac{\vec{v}}{|\vec{v}|} \quad \text{versore di } \vec{v} \text{ (vettore)}$$

$$|\hat{v}| = \frac{|\vec{v}|}{|\vec{v}|} = 1 \quad \text{modulo di } \hat{v}$$



$$\vec{v} = v_x \hat{x} + v_y \hat{y} + v_z \hat{z} \rightarrow \text{modo alternativo di scrivere il vettore}$$

$$\vec{v}_1 = (v_{1x}, v_{1y}, v_{1z})$$

$$\vec{v}_2 = (v_{2x}, v_{2y}, v_{2z})$$

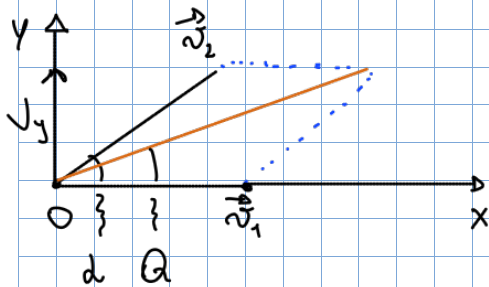
$$\vec{v} = \vec{v}_1 + \vec{v}_2 = (v_{1x} + v_{2x}, v_{1y} + v_{2y}, v_{1z} + v_{2z})$$

Es

$$\vec{v}_1 = (5, \sqrt{2}, 0)$$

$$\vec{v}_2 = (0, 4, 5)$$

$$\vec{v} = (5, 4 + \sqrt{2}, 5)$$



$$\vec{v}_1 = (v_1, 0, 0)$$

$$\vec{v}_2 = (v_2 \cos \alpha, v_2 \sin \alpha, 0)$$

$$\vec{v} = (v_1 + v_2 \cos \alpha, v_2 \sin \alpha, 0)$$

$$v_y = v \sin \alpha = v_2 \sin \alpha$$

$\Downarrow$

$$\sin \alpha = \frac{v_2}{v} \sin \alpha$$

$$V \cos \alpha = v_1 + v_2 \cos \alpha \Rightarrow \cos \alpha = \frac{v_1 + v_2 \cos \alpha}{V}$$

$$\cos \alpha = \sqrt{1 - \sin^2 \alpha} = \sqrt{1 - \frac{v_2^2}{V^2} \sin^2 \alpha} = \frac{1}{V} \sqrt{V^2 - v_2^2 \sin^2 \alpha} = \dots$$

$$V = \sqrt{(v_1 + v_2 \cos \alpha)^2 + v_2^2 \sin^2 \alpha} \Rightarrow V^2 = v_1^2 + v_2^2 \cos^2 \alpha + 2v_1 v_2 \cos \alpha + v_2^2 \sin^2 \alpha$$

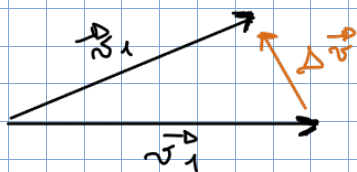
$$\dots = \frac{1}{V} \sqrt{v_1^2 + v_2^2 \cos^2 \alpha + 2v_1 v_2 \cos \alpha + \cancel{v_2^2 \sin^2 \alpha} - \cancel{v_2^2 \sin^2 \alpha}}$$

$$= \frac{1}{V} \sqrt{(v_1 + v_2 \cos \alpha)^2} = \frac{v_1 + v_2 \cos \alpha}{V}$$

$$\vec{v}_1 = (v_{1x}, v_{1y}, v_{1z})$$

$$\vec{v}_2 = (v_{2x}, v_{2y}, v_{2z})$$

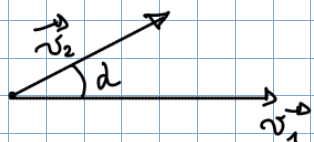
$$\Delta \vec{v} = \vec{v}_2 - \vec{v}_1 = (v_{2x} - v_{1x}, v_{2y} - v_{1y}, v_{2z} - v_{1z}) \quad \text{differenza tra vettori}$$



$$\vec{v}_1 = (v_{1x}, v_{1y}, v_{1z})$$

$$\vec{v}_2 = (v_{2x}, v_{2y}, v_{2z})$$

$$\vec{v}_1 \cdot \vec{v}_2 = (v_{1x} \cdot v_{2x}, v_{1y} \cdot v_{2y}, v_{1z} \cdot v_{2z}) \quad \text{prodotto tra vettori}$$



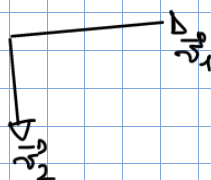
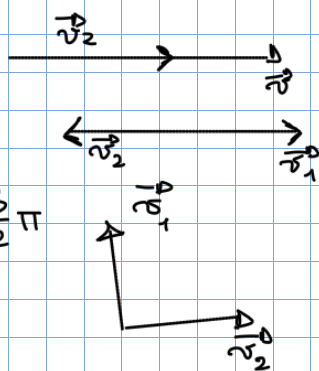
$$\vec{v}_1 \cdot \vec{v}_2 = v_1 v_2 \cos \alpha$$

prodotto alternativo tra vettori

$$\cos \alpha = 1 \Rightarrow \alpha = 0$$

$$\cos \alpha = -1 \Rightarrow \alpha = \pi$$

$$\cos \alpha = 0 \Rightarrow \alpha = \frac{\pi}{2}, \frac{3}{2}\pi$$



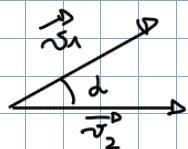
non ho capito
!!

$$\vec{v}_1 = (v_{1x}, v_{1y}, v_{1z})$$

$$\vec{v}_2 = (v_{2x}, v_{2y}, v_{2z})$$

$$|\vec{v}_1 \times \vec{v}_2| = v_1 v_2 \sin \alpha$$

prodotto vettoriale



$$\vec{v}_1 \times \vec{v}_2 = - \vec{v}_2 \times \vec{v}_1$$

$$\vec{v}_1 \times \vec{v}_2 = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ v_{1x} & v_{1y} & v_{1z} \\ v_{2x} & v_{2y} & v_{2z} \end{vmatrix} = \hat{x} (v_{1y} v_{2z} - v_{2y} v_{1z}) + \hat{y} (\dots) + \hat{z} (\dots)$$